Nahid Hasan Sagar

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PROFESSIONAL SUMMARY

M.S. candidate in Statistical Science at Indiana University Bloomington with experience in computational statistics, reproducible data analysis, and benchmarking research workflows. Applies R, Python, C++, MATLAB, and Bash to support quantitative research, data quality checks, and efficient analysis pipelines for real-world data.

EDUCATION

Indiana University, College of Arts and Sciences Bloomington, IN
Master of Science in Statistical Science *Expected May 2027*
Thesis: Time-Series Analysis Using Bayesian Additive Regression Trees (BART)
Graduate Coursework: Introduction to Statistical Computing; Applied Statistical Computing; Fundamentals of Statistics I–II; Applied Linear Models I–II
Current Coursework: Statistical Consulting; Research in Statistics; Introduction to Analysis I

University of Dhaka Dhaka, Bangladesh
Bachelor of Science in Physics *August 2024*

EXPERIENCE

Research Assistant, StatMIND Lab 2025 – Present
Indiana University Bloomington *Bloomington, IN*
- Develop computational methods to improve the efficiency and scalability of Bayesian neuroimaging pipelines using R and C++.
- Achieved a 4 to 5× speedup in Bayesian neuroimaging algorithms and benchmarked performance using simulated and real fMRI data.
- Build reproducible benchmarking workflows to compare runtime, scalability, and computational performance across implementations.
- Support research analysis by organizing outputs, checking consistency across runs, and presenting results for technical discussion.

Teaching Assistant, Department of Statistics 2025 – Present
Indiana University Bloomington *Bloomington, IN*
- Support undergraduate Statistics courses through classroom instruction, discussion sections, and individualized guidance.
- Evaluate approximately 200 assignments per grading cycle and provide detailed feedback on statistical reasoning, computation, and interpretation.
- Conduct weekly office hours and exam-review sessions to strengthen students' conceptual understanding and problem-solving skills.
- Communicate technical concepts clearly to students with varied backgrounds while maintaining a high standard of quality and professionalism.

Senior Mathematics Instructor 2016 – 2025
Udvash Academic & Admission Care *Dhaka, Bangladesh*
- Taught and mentored more than 100,000 students through classroom instruction and national online learning platforms.
- Prepared students for competitive university admission examinations through structured lessons, practice materials, and academic planning.
- Translated advanced mathematical concepts into accessible problem-solving strategies for students with varied educational backgrounds.
- Built strong one-on-one communication skills through sustained mentoring, feedback, and instructional support.

SELECTED RESEARCH & PROJECTS

M.S. Thesis: Time-Series Analysis Using BART *R, Bayesian Modeling*
- Conduct thesis research on the use of Bayesian Additive Regression Trees for time-series analysis under the supervision of Dr. Matthew T. Pratola.
- Apply statistical modeling and reproducible analysis workflows to evaluate predictive performance on time-series data.

Ordinary Least Squares from Scratch *Python, NumPy, Pandas, Matplotlib*
- Develop an educational implementation of ordinary least squares regression from first principles, emphasizing matrix-based estimation and transparent statistical computation.
- Validate outputs with careful numerical checks and clear visualization of model fit and residual behavior.

TECHNICAL SKILLS

Statistical Methods: Bayesian inference, BART, MCMC, hierarchical modeling, time-series analysis, linear and logistic regression, mixed-effects models, bootstrap methods, experimental design, model selection, and predictive model evaluation
Research and Analysis: Benchmarking, reproducible analysis workflows, statistical reporting, visualization, and literature review
Programming: R, Python, C, C++, MATLAB, Bash, HTML, CSS, and JavaScript
Computing and Tools: Unix/Linux, numerical optimization, algorithm benchmarking, code profiling, Git, GitHub, LaTeX, RStudio, and VS Code
Languages: Bengali (native), English (fluent), Hindi (conversational), Urdu (conversational)

LEADERSHIP & HONORS

- Attended the 2026 Velocity Conference at UC Berkeley as an Indiana University representative for the science community.
- Collaborated on the development of a comprehensive online Linear Algebra curriculum during the COVID-19 pandemic.
- Co-organized the Australian Mathematics Competition in Bangladesh in 2014.